

KUMAR VARENYA

(604) 727-7638 | kumarvarenya85@gmail.com

kvaranya.github.io | [LinkedIn](#) | [GitHub](#) | [ORCID](#) | Canadian Permanent Resident

TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL, R, Bash, SystemVerilog, VHDL, ARM Assembly
Embedded & Hardware: Intel DE1-SoC, Intel Quartus, Altium Designer, ModelSim, Arduino, Raspberry Pi, LTspice
Libraries & Frameworks: Pandas, Scikit-learn, Matplotlib, Tidyverse, tidymodels, Flask, PostGIS, Leaflet.js
Developer Tools: Git, Linux/Unix, Docker, CMake, Make, GCC, GDB, Valgrind, n8n, Claude Code Hooks, Cursor
Databases & APIs: PostgreSQL, Airtable, REST APIs, JSON, Jupyter Notebooks, Tableau, Excel

EXPERIENCE

Meta (Reality Labs)

Toronto, ON

Software Engineer

April 2026 – Present

- Built and optimized C++ perception pipelines for mixed-reality prototypes, fusing camera and IMU data for real-time 6DoF tracking and spatial mapping while reducing end-to-end frame latency by 18%.
- Developed Python/C++ evaluation tooling and automated regression tests for computer vision and sensor-fusion models, cutting experiment turnaround by 35% and surfacing tracking regressions before device deployment.
- Partnered with research, graphics, and systems engineers to productionize prototype XR features, profiling CPU/GPU bottlenecks and memory usage to improve frame consistency under on-device power and thermal constraints.

UBC Investment Management Trust Inc.

Vancouver, BC

Risk/Systems Engineer (UBC CS DMM: Operations Research & Financial Engineering)

September 2025 – April 2026

- Developed quantitative optimization models for allocation across multi-asset \$2.336 billion endowment portfolio.
- Engineered risk analytics infrastructure integrating factor models, scenario simulations, and stress-testing.
- Collaborated with UBCiM to translate operational research outputs into actionable allocation and hedging strategies.

Amazon

Vancouver, BC

Software Development Engineer Intern (AWS)

May 2024 – December 2024

- Built AI software package using Docker, Kubernetes, and AWS (CDK, Fargate, Bedrock), improving efficiency by 50%.
- Designed secure LLM prompt templates and PII detection, cutting errors by 70% with zero breaches.
- Implemented unit and integration tests (100% pass rate); contributed to key feature deployment for Prime Day.

Population Data BC

Vancouver, BC

Software Programmer

September 2022 – December 2025

- Built Python and SQL ETL pipelines to boost data throughput and reduce latency for large epidemiological datasets.
- Developed containerized microservices with API gateways to enable secure, compliant data access.
- Refactored legacy code using design patterns and optimizations to improve scalability and maintainability.

University of British Columbia, Department of Computer Science

Vancouver, BC

Teaching Assistant (CPSC 110, CPSC 121)

January 2023 – April 2024

- Led weekly tutorials for 2000+ students, improving understanding of functional programming and discrete math.
- Graded over 500 assignments and exams per week and provided constructive feedback to improve student performance.
- Collaborated with course instructors to update course materials, aligning content with current programming practices.

Oil and Natural Gas Corporation

New Delhi, IN

Summer Intern

Summer 2022, Summer 2023

- Analyzed pipeline monitoring data to identify potential failures, reducing unplanned maintenance incidents by 20%.
- Supported the implementation of GIS mapping tools for resource allocation and logistics planning.
- Assisted in designing database systems to track over 1,000 equipment inspections and maintenance schedules.

RESEARCH EXPERIENCE

Carnegie Mellon University

Pittsburgh, PA

Undergraduate Research Assistant

May 2025 – July 2025

- Investigating compositional verification techniques for concurrent software systems to enhance modular analysis.
- Applied static analysis to identify data races and synchronization bugs in multithreaded C/C++ programs.

UBC Pacific Laboratory of Artificial Intelligence

Vancouver, BC

Data Engineering Research Assistant

September 2023 – April 2024

- Collected, curated, and annotated Minecraft gameplay datasets to support behavioral and AI-driven analysis.
- Built preprocessing pipelines to structure spatial-temporal data for downstream machine learning tasks.

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Computer Engineering & Honours Mathematics

2021 – 2026